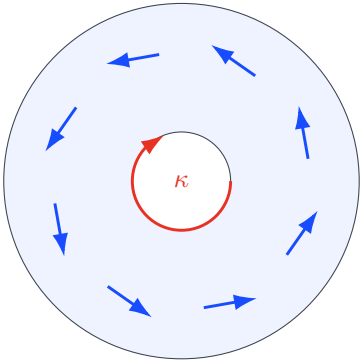
普通旋度与集中旋度：穿孔后由内边界保留轴线贡献



外边界逆时针，内边界顺时针

$$\rho > 0: \quad \nabla \times \mathbf{A} = 2a\,\mathbf{e}_z$$

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = 2\pi a R^2 + \kappa$$

$$\nabla \times \mathbf{A} = 2a\mathbf{e}_z + \kappa\delta(x)\delta(y)\mathbf{e}_z$$

箭头示意 $a, \kappa > 0$；通式允许两项系数为任意符号。